



Here you can see the incredible fluid dynamics that the PhysX PPU makes possible. The picture shows soap bubbles on a car being washed—realistically!

that we're interested in right now, and the PPU is surely the most phenomenal thing for PC gamers since the coming of age of the GPU!

What Will Improve?

In order to better understand the differences between the physics computations that a really high-end CPU can achieve, and what a PhysX-powered system can achieve, head to <http://physx.ageia.com/footage.html> and www.ageia.com/standalone_pages/ageia_physx.aspx. The difference between an explosion now and after using PhysX is:

Normal: "Boom, clang, crash, poof (smoke clears and guy standing more than 10 metres away from explosion doesn't even blink)"

PhysX: "Kaboom, clang, crash, clang (x100), crash (x50), poof, (smoke billows, and people standing 20 metres away, as well as you yourself, get a little damage due to the flying shrapnel)"

This means you cannot run around a map carrying a rocket launcher, because you'll only succeed in blowing yourself up—but who said you could do that in real life anyway? Realism is the key, and PPU solutions will soon have us believing. For one, PhysX promises more realistic explosions, where things that explode actually do so into thousands of pieces—and all of them random-sized of course, not like dividing a truck into a thousand equally-sized pieces and then exploding it! There will be interactive environments, so you can actually destroy a building if you have enough firepower, or at least make a hole in a wall. Everything in a map could be made destructible if the game engineers so desire. And, of course, when you're destroying stuff, explosions, fire and gas will behave like in the real world.

Apart from the object interactions we're raving about, PhysX will also mean more realistic fluid dynamics—so oil will look and flow like oil, and smoke will *not* travel through a wall or pillar, but behave more realistic. Many of you would have noticed that, in games, smoke

behaves the same way outdoors as it does indoors. Sometimes a game's physics is so ridiculous that it will depict trees and grass waving in a strong breeze, yet the vehicle you just blew up will send up an absolutely straight column of smoke that doesn't disperse. Also, when indoors, you would have seen smoke actually going through a door and just disappear when it hits the ceiling; in reality, smoke would collect at the ceiling and then spread outwards, slowly engulfing the whole room in a black, acrid fog. PhysX promises experiences like these in games that will be developed using their software and hardware technology.

But What About ATI And NVIDIA?

In any talk about gaming, you cannot forget about the two graphics giants. Both companies seem to be convinced that they will soon make the need for a separate PPU, GPU and CPU combination redundant. Both feel their GPUs are powerful enough to do physics computations, and that it's just a matter of utilising all the gigaflops of processing power that are not used on their latest-generation cards.

NVIDIA has tied up with Havoc, a company that has done physics simulation work for some of the biggest game developers in the world, to create the Havoc FX Engine. This engine is basically a software solution that will command the CPU to send physics processing load to the GPU. According to reports, the FX engine will work with both single as well as multiple NVIDIA GPU-based systems (SLI). However, the cards that will support running this software will have to be 7600 or higher series cards. Now this isn't ideal, because the majority of people using NVIDIA's products own 6-series cards (such as the 6600GT). Of course, even with the lower penetration of NVIDIA's 7600 series and above cards (7900 recommended), they're still a lot better off than Ageia, who are starting from naught!

According to NVIDIA, even using a single high-end 7-series card to do your physics processing will increase the amount of physics processed. In terms of evidence, they provided a fixed amount of physics processing (15,000 boulders colliding). They read frame rates for when the CPU did the calculations, and compared it to the GPU taking over the physics calculations. The results sound too good to be true: the CPU managed 6.2 fps while the GPU managed 64.5 fps! The same GPU pair (for SLI mode) was used in both tests, so it wasn't a with-and-without-GPU comparison!

If NVIDIA can reproduce this success with physics processing in real world tests, Ageia may already be in trouble. With almost every NVIDIA-loyal gamer buying (or at least saving for) an SLI pair of cards, a lot of people are going to have a lot of free computing space on their GPUs which, if Havoc FX does what it promises, will eliminate the need for a separate PPU!

ATI are convinced that their Crossfire solution will soon have both NVIDIA and Ageia licking their wounds. According to them, their latest brands of graphics chipsets already have physics processing capabilities. They're already working on an API that will give a game direct access to their GPUs, thus significantly reduc-